SDIE VERSION 5 – DEEPSEEK STRUCTURAL REASONING ENGINE

MISSION

Redesign SDIE into a Structural Reasoning Engine capable of understanding structural drawings the same way an experienced structural estimator interprets them.

Version 4 improved classification accuracy through:

- Component Atlas
- Layer Profiles
- DeepSeek Ambiguity Resolution
- Structural Graphs
- Knowledge Base

However classification accuracy remains insufficient because the system still relies heavily on layer interpretation and geometry heuristics.

Version 5 must introduce estimator-style reasoning.

The system must learn:

- How estimators identify structural components
- How estimators use drawing context
- How estimators interpret layers
- How estimators interpret annotations
- How estimators resolve ambiguity
- How estimators separate slab regions from exclusions

The objective is not slab detection.

The objective is structural understanding.

PRIMARY OBJECTIVE

Increase component classification accuracy from 50–60% toward 85–95%.

Primary target classes:

- Slab

- Beam
- Column
- Shear Wall

Supporting classes:

- Structural Wall
- Lift Core
- Stair Core
- Shaft
- Opening
- Unknown

Business quantity engines remain:

- Slab Calculator
- Beam Calculator
- Column Calculator
- Shear Wall Calculator

All other classes support exclusions and diagnostics.

PROMPT ENGINEERING FRAMEWORK (NEW)

Every DeepSeek request MUST include the following context.

Scale Context

Always pass drawing scale.

Examples:

Scale = 1:50

Scale = 1:100

Scale = 1:200

DeepSeek must never assume scale.

Unit Context

Always pass unit system.

Examples:

Units = Millimeters

Units = Meters

Units = Feet

Units = Inches

Units = Imperial

Units = Metric

DeepSeek must never assume units.

Geometry interpretation must depend on the supplied unit system.

Layer Context Injection

Always provide available layer names.

Example:

Layers Found:

WALL WALLS COLUMN COLUMNS COL S-COLS S-COL HATCH BEAM S-BEAM STR-BEAM SLAB A-FLOR-IDEN SHEARWALL S-SHEARWALL CORE STAIR OPENING GRID TEXT DIMENSION

DeepSeek should treat layer names as strong evidence.

Layer names are not absolute truth.

Geometry Context Injection

Every candidate entity sent to DeepSeek must include:

- Entity Type
- Area
- Perimeter
- Aspect Ratio
- Bounding Box
- Connected Beams
- Connected Columns

- Connected Walls
- Neighbour Relationships

Example:

{ “entity_id”: “ENT_001”, “entity_type”: “LWPOLYLINE”, “area”: 35.6, “perimeter”: 24.8, “aspect_ratio”: 7.4, “connected_beams”: 4, “connected_columns”: 2 }

Annotation Context Injection

Provide nearby text.

Examples:

THK 150

THK 200

THK 250

B12

C5

SW1

LIFT

STAIR

OPENING

SHAFT

Annotations are strong semantic indicators.

Cursor Context Injection (NEW)

Always include:

@projects_manifest.json

@data/layer_profiles.json

@data/atlas/component_atlas.json

@data/knowledge_base/structural_kb.json

@src/sdie/config.py

@src/sdie/classification/

@src/sdie/rag/

@src/sdie/validation/

DEEPSEEK SYSTEM ROLE

Every DeepSeek request must begin with:

“You are a senior structural quantity surveyor, structural estimator, and CAD interpretation specialist.

Your responsibility is structural component identification.

You must classify structural entities using:

1. Layer names
2. Geometry
3. Annotation text
4. Structural topology
5. Historical estimator examples
6. Knowledge base context
7. Atlas examples

You are forbidden from calculating:

- Area
- Volume
- Concrete
- Shuttering

You are only responsible for structural understanding.

When layers are missing, infer component types from geometry, topology, and estimator patterns.

Return confidence scores for every entity.

Flag uncertain entities for review.”

MULTI-EVIDENCE CLASSIFICATION ENGINE

DeepSeek must never classify using a single signal.

Classification must combine:

Layer Information = 25%

Geometry Features = 25%

Topology Features = 20%

Annotation Features = 15%

Atlas Similarity = 10%

Knowledge Base Similarity = 5%

Classification must be evidence-based.

GEOMETRY FALLBACK REASONING (NEW)

If layers are unnamed, incorrect, generic, or ambiguous:

DeepSeek must infer component type from geometry.

Beam Indicators

- Long narrow geometry
- High aspect ratio
- Connects columns
- Repeated pattern
- Structural alignment

Column Indicators

- Small rectangular geometry
- Grid aligned
- Supports beams
- Repeating structural pattern

Shear Wall Indicators

- Continuous wall geometry
- Large thickness
- Core proximity

- Structural continuity

Slab Indicators

- Large enclosed framed region
- Surrounded by beams
- Thickness annotation nearby
- Not classified as wall/core/opening

Lift Core Indicators

- Core enclosure
- Lift annotation
- Wall enclosure geometry

Stair Core Indicators

- Stair annotation
- Flight geometry
- Core adjacency

FEW SHOT ESTIMATOR REASONING

Before classification retrieve:

Top 5 similar Atlas samples.

Top 5 similar Knowledge Base samples.

Examples:

Project: INIZIO

Layer: S-BEAM

Geometry:

Length = 12.4m

Width = 0.3m

Classification:

Beam

Confidence:

100%

Use nearest estimator examples before classification.

RAG RETRIEVAL STRATEGY

Retrieve by:

1. Layer Match
2. Annotation Match
3. Geometry Similarity
4. Project Similarity
5. Structural Relationship Similarity

Combine results before DeepSeek reasoning.

ENTITY LEVEL CONFIDENCE (NEW)

Every classified entity must return:

{ “entity_id”: “ENT_001”, “classification”: “Beam”, “confidence”: 94, “review_required”: false, “evidence”: { “layer_match”: 100, “geometry_match”: 95, “annotation_match”: 80, “atlas_match”: 92 } }

CONFIDENCE THRESHOLDS

Confidence ≥ 90

Automatic Acceptance

Confidence 75–89

Accepted With Warning

Confidence < 75

Manual Review Required

Confidence < 60

Force Validation Queue

STRUCTURAL EXCLUSION REASONING

Before slab identification explicitly classify and exclude:

- Beam
- Column
- Shear Wall
- Structural Wall
- Lift Core
- Stair Core
- Shaft
- Opening

Only after exclusions:

Remaining Framed Region

=

Potential Slab

No slab may be generated before exclusion analysis.

AMBIGUITY REVIEW ENGINE

Create Review Queue:

review_queue.json

Store:

- Entity ID
- Proposed Classification
- Confidence
- Evidence
- Suggested Alternatives

Example:

{ “entity_id”: “ENT_422”, “classification”: “Beam”, “confidence”: 63, “alternatives”: [“Shear Wall”, “Structural Wall”] }

HISTORICAL PROJECT LEARNING

Projects:

INIZIO

TRUST_OFFICE

Future:

Project4

Project5

Every completed project contributes:

- Atlas Samples
- Layer Profiles
- Structural Patterns
- Annotation Patterns
- Quantity Relationships

Knowledge accumulates permanently.

CLASSIFICATION TARGETS

Version 5 Target:

Beam Accuracy ≥ 90%

Column Accuracy ≥ 90%

Shear Wall Accuracy ≥ 90%

Slab Accuracy ≥ 90%

Overall Classification ≥ 90%

Low confidence entities ≤ 5%

FINAL DIRECTIVE

Do not improve slab detection directly.

Improve structural reasoning.

Teach the system how estimators think.

DeepSeek is a structural reasoning engine.

The quantity engine remains deterministic.

All quantity estimation must occur only after:

1. Component Classification
2. Structural Graph Generation
3. Semantic Building Model Creation
4. Structural Exclusion Analysis

The objective is not slab detection.

The objective is structural understanding.